

Q. 9. How is a direct-connected exciter arranged in an alternator ?

Ans. The armature of the exciter is mounted on the shaft of the alternator close to the spider hub. In some cases, it is mounted at a distance sufficient to permit a pedestal and bearing to be placed between the exciter and the hub.

Q. 10. Any advantage of a direct-connected exciter ?

Ans. Yes, economy of space.

Q. 11. Any disadvantage ?

Ans. The exciter has to run at the same speed as the alternator which is slower than desirable. Hence, it must be larger for a given output than the gear-driven type, because it can be run at high speed and so made proportionately smaller.

OBJECTIVE TESTS – 37

- The frequency of voltage generated by an alternator having 4-poles and rotating at 1800 r.p.m. is hertz.
(a) 60 (b) 7200
(c) 120 (d) 450.
- A 50-Hz alternator will run at the greatest possible speed if it is wound for poles.
(a) 8 (b) 6
(c) 4 (d) 2.
- The main disadvantage of using short-pitch winding in alternators is that it
(a) reduces harmonics in the generated voltage
(b) reduces the total voltage around the armature coils
(c) produces asymmetry in the three phase windings
(d) increases Cu of end connections.
- Three-phase alternators are invariably Y-connected because
(a) magnetic losses are minimised
(b) less turns of wire are required
(c) smaller conductors can be used
(d) higher terminal voltage is obtained.
- The winding of a 4-pole alternator having 36 slots and a coil span of 1 to 8 is short-pitched by degrees.
(a) 140 (b) 80
(c) 20 (d) 40.
- If an alternator winding has a fractional pitch of $5/6$, the coil span is degrees.
(a) 300 (b) 150
(c) 30 (d) 60.
- The harmonic which would be totally eliminated from the alternator e.m.f. using a fractional pitch of $4/5$ is
(a) 3rd (b) 7th
(c) 5th (d) 9th.
- For eliminating 7th harmonic from the e.m.f. wave of an alternator, the fractional-pitch must be
(a) $2/3$ (b) $5/6$
(c) $7/8$ (d) $6/7$.
- If, in an alternator, chording angle for fundamental flux wave is α , its value for 5th harmonic is
(a) 5α (b) $\alpha/5$
(c) 25α (d) $\alpha/25$.
- Regarding distribution factor of an armature winding of an alternator which statement is false?
(a) it decreases as the distribution of coils (slots/pole) increases
(b) higher its value, higher the induced e.m.f. per phase
(c) it is not affected by the type of winding either lap, or wave
(d) it is not affected by the number of turns per coil.
- When speed of an alternator is changed from 3600 r.p.m. to 1800 r.p.m., the generated e.m.f./phases will become
(a) one-half (b) twice
(c) four times (d) one-fourth.
- The magnitude of the three voltage drops in an alternator due to armature resistance, leakage reactance and armature reaction is solely determined by
(a) load current, I_a

- (b) p.f. of the load
(c) whether it is a lagging or leading p.f. load
(d) field construction of the alternator.
13. Armature reaction in an alternator primarily affects
(a) rotor speed
(b) terminal voltage per phase
(c) frequency of armature current
(d) generated voltage per phase.
14. Under no-load condition, power drawn by the prime mover of an alternator goes to
(a) produce induced e.m.f. in armature winding
(b) meet no-load losses
(c) produce power in the armature
(d) meet Cu losses both in armature and rotor windings.
15. As load p.f. of an alternator becomes more leading, the value of generated voltage required to give rated terminal voltage
(a) increases
(b) remains unchanged
(c) decreases
(d) varies with rotor speed.
16. With a load p.f. of unity, the effect of armature reaction on the main-field flux of an alternator is
(a) distortional (b) magnetising
(c) demagnetising (d) nominal.
17. At lagging loads, armature reaction in an alternator is
(a) cross-magnetising (b) demagnetising
(c) non-effective (d) magnetising.
18. At leading p.f., the armature flux in an alternator the rotor flux.
(a) opposes (b) aids
(c) distorts (d) does not affect.
19. The voltage regulation of an alternator having 0.75 leading p.f. load, no-load induced e.m.f. of 2400V and rated terminal voltage of 3000V is percent.
(a) 20 (b) -20
(c) 150 (d) -26.7
20. If, in a 3- ϕ alternator, a field current of 50A produces a full-load armature current of 200 A on short-circuit and 1730 V on open circuit, then its synchronous impedance is ohm.
(a) 8.66 (b) 4
(c) 5 (d) 34.6
21. The power factor of an alternator is determined by its
(a) speed
(b) load
(c) excitation
(d) prime mover.
22. For proper parallel operation, a.c. polyphase alternators must have the same
(a) speed (b) voltage rating
(c) kVA rating (d) excitation.
23. Of the following conditions, the one which does not have to be met by alternators working in parallel is
(a) terminal voltage of each machine must be the same
(b) the machines must have the same phase rotation
(c) the machines must operate at the same frequency
(d) the machines must have equal ratings.
24. After wiring up two 3- ϕ alternators, you checked their frequency and voltage and found them to be equal. Before connecting them in parallel, you would
(a) check turbine speed
(b) check phase rotation
(c) lubricate everything
(d) check steam pressure.
25. Zero power factor method of an alternator is used to find its
(a) efficiency
(b) voltage regulation
(c) armature resistance
(d) synchronous impedance.
26. Some engineers prefer 'lamps bright' synchronization to 'lamps dark' synchronization because
(a) brightness of lamps can be judged easily
(b) it gives sharper and more accurate synchronization
(c) flicker is more pronounced
(d) it can be performed quickly.
27. It is never advisable to connect a stationary alternator to live bus-bars because it
(a) is likely to run as synchronous motor
(b) will get short-circuited
(c) will decrease bus-bar voltage though momentarily
(d) will disturb generated e.m.fs. of other alternators connected in parallel.

28. Two identical alternators are running in parallel and carry equal loads. If excitation of one alternator is increased without changing its steam supply, then
- it will keep supplying almost the same load
 - kVAR supplied by it would decrease
 - its p.f. will increase
 - kVA supplied by it would decrease.
29. Keeping its excitation constant, if steam supply of an alternator running in parallel with another identical alternator is increased, then
- it would over-run the other alternator
 - its rotor will fall back in phase with respect to the other machine
 - it will supply greater portion of the load
 - its power factor would be decreased.
30. The load sharing between two steam-driven alternators operating in parallel may be adjusted by varying the
- field strengths of the alternators
 - power factors of the alternators
 - steam supply to their prime movers
 - speed of the alternators.
31. Squirrel-cage bars placed in the rotor pole faces of an alternator help reduce hunting
- above synchronous speed only
 - below synchronous speed only
 - above and below synchronous speeds both
 - none of the above.
- (Elect. Machines, A.M.I.E. Sec. B, 1993)
32. For a machine on infinite bus active power can be varied by
- changing field excitation
 - changing of prime mover speed
 - both (a) and (b) above
 - none of the above.
- (Elect. Machines, A.M.I.E. Sec. B, 1993)

ANSWERS

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| 1. a | 2. d | 3. b | 4. d | 5. d | 6. b | 7. c | 8. d | 9. a | 10. b | 11. a |
| 12. a | 13. d | 14. b | 15. c | 16. a | 17. d | 18. b | 19. b | 20. c | 21. b | 22. b |
| 23. d | 24. b | 25. b | 26. b | 27. b | 28. a | 29. c | 30. c | 31. c | 32. b | |